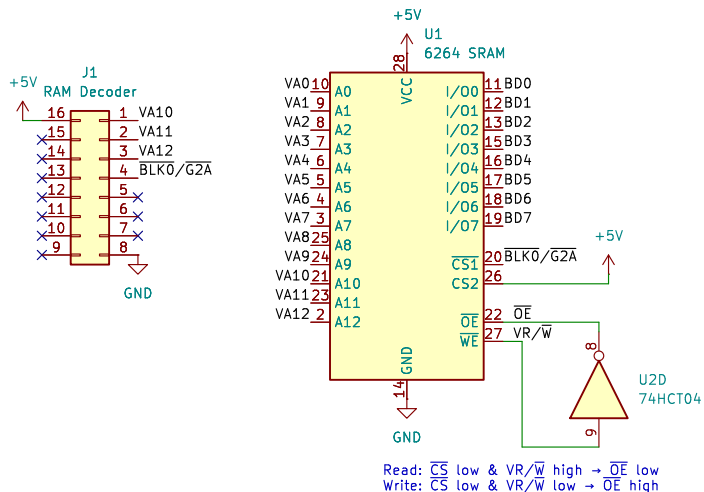
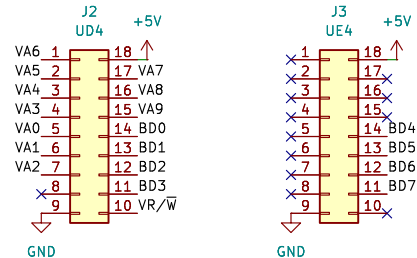


Signals to/from RAM decoder (74LS138)
Original decoder activates internal RAM only for $\overline{RAM0}$ and $\overline{RAM4}$ to $\overline{RAM5}$
We want to activate our replacement for every 1KB chunk in BLK0, including $\overline{RAM1}$ to $\overline{RAM3}$
This will give us the full 8KB in BLK0 – standard 5KB plus 3KB RAM expansion
Greatly simplified design:
We essentially only need the $\overline{BLK0}$ select signal, used as an enable on the original 74LS138.
So I think we just need to use that same signal as our RAM select ... ?

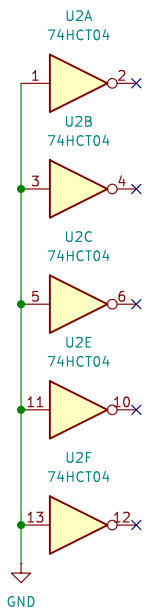
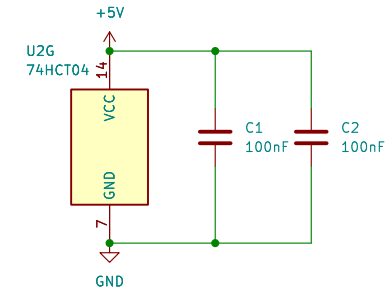


Original 2114 RAM chips are removed from the VIC.
Two are replaced with IC sockets so daughterboard can be installed.



Refer to the "DIP-16 Breakout Board" for the J1 connection

Power & Decoupling



Long words:
[FI] Lentokonesuihkuturbiinimootoriarpumekaanikoaliupseerioppilas
[EN] "Airplane jet turbine engine auxiliary mechanic non-commissioned officer student" (an old military training rank).
[RU] Тетрагидропиранилициклопентилтетрагидропиридопиридиновые
[EN] Tetrahydropyranylcyclopentyltetrahydropyridopyridine (class of organic compounds)
[PL] Dziewięćsetdziesięćdziesięćdziesięćdziesięćnarodowościowego
[EN] Genitive singular form of "of nine-hundred-and-ninety-nine nationalities."
[DE] Rinderkennzeichnungsfleischetikettierungsüberwachungsaufgabenübertragungsgesetz
[EN] "Law on the delegation of duties for the supervision of cattle identification and beef labeling."
[JP] 虎が人を噛もうとするときうなり声
[EN] "The growl a tiger makes when it's about to bite a person."

U2	74HCT04_Hex Inverter
U1	6264 SRAM
C1, C2	100nF Decoupling Capacitor
J1	2x8 Dupont Female-Female Cables to VIC
J2 & J3	Pair of 1x9 SIL Pin Headers or 2x9 Wirewrap Sockets

www.youtube.com/@Brfff

Brett Hallen

Sheet: /

File: VIC-20_Internal_8KB_RAM_Simple.kicad_sch

Title: Commodore VIC-20 Simple BLK0 RAM Expansion

Size: A4

Date: 28/MAY/2026

Rev: A

KiCad E.D.A. 9.0.9

Id: 1/1